

OMAR AHMED ALI

Computer Engineering Student

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PROFESSIONAL SUMMARY

Computer Engineering student (Semester 7, GPA 3.7) with hands-on experience building and deploying full backend systems using Node.js, Express, and MySQL. Skilled in REST API design, relational database modeling (including many-to-many relationships), authentication and authorization (JWT, bcrypt), and cloud deployment (Railway). Also experienced in frontend development with HTML, CSS, and JavaScript. Currently seeking a Software Engineering internship or freelance backend projects.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C++, C, C#, Assembly (8086/8051)

Backend: Node.js, Express.js, REST APIs, JWT Authentication, bcrypt, mysql2/promise

Databases: MySQL — relational schema design, JOINS, foreign keys, many-to-many relationships

Frontend: HTML, CSS, JavaScript

BACKEND PROJECTS

Small Student Portal — Deployed Live — *Node.js, Express, MySQL, JWT*

Repo: github.com/omarrrahmeddd/small-student-portal

- Built and deployed a full REST API for student course enrollment, live on a public URL with a cloud-hosted MySQL database (Railway).
- Designed a many-to-many relational schema (students ↔ courses via an enrollments join table) with SQL JOIN queries.
- Implemented signup/login with bcrypt password hashing and JWT-based authentication middleware protecting enrollment and course-lookup routes.
- Debugged and resolved a real production deployment issue (database connection configuration across environments), demonstrating practical troubleshooting skills.

EMBEDDED SYSTEMS PROJECTS

RFID-Based Access Control System — *8051 (AT89C51), Arduino, Embedded C/Assembly, Proteus*

- Designed a two-processor access control system: an Arduino + MFRC522 module handles RFID scanning over SPI and authorization logic, while an 8051 (AT89C51) microcontroller drives all outputs (LCD, servo-driven lock, LEDs) via a serial (UART) link between the two.
- Implemented auto-enrollment so the system learns and remembers valid cards across sessions, with non-volatile storage of card IDs in EEPROM.
- Designed and simulated the full circuit in Proteus prior to hardware assembly, then built and debugged the physical circuit end-to-end.

OTHER PROJECTS

Huffman File Compression — *C++*

- Developed a file compression/decompression tool using the Huffman Coding algorithm with binary trees and priority queues.

Personal Portfolio — *HTML, CSS, JavaScript*

- Designed and deployed a responsive personal website showcasing technical projects, hosted on Vercel.

Internet Connection Checker — *JavaScript*

- Built a real-time utility to monitor network connectivity using DOM manipulation and async logic.

EDUCATION

Bachelor of Computer Engineering — Semester 7

Arab Academy for Science, Technology & Maritime Transport (AASTMT) | Expected Graduation: 2027

GPA: 3.7 / 4.0

RELEVANT COURSEWORK

- Data Structures & Algorithms — Trees, Graphs, Hash Tables, efficient problem solving in C++ and Java
- Object-Oriented Programming — Inheritance, Polymorphism, applied across multiple software projects
- Database Management Systems — Relational design, SQL, normalization
- Microprocessors — 8086/8088 and 8051 assembly, hardware interfacing
- Data Analytics & Machine Learning using Python — Regression, Classification (SVM, Naive Bayes, Logistic Regression), Clustering (K-Means, GMM), and text/NLP (TF-IDF)